

ICA

Blind Source Separation

Pharma Batch Quality Engineering

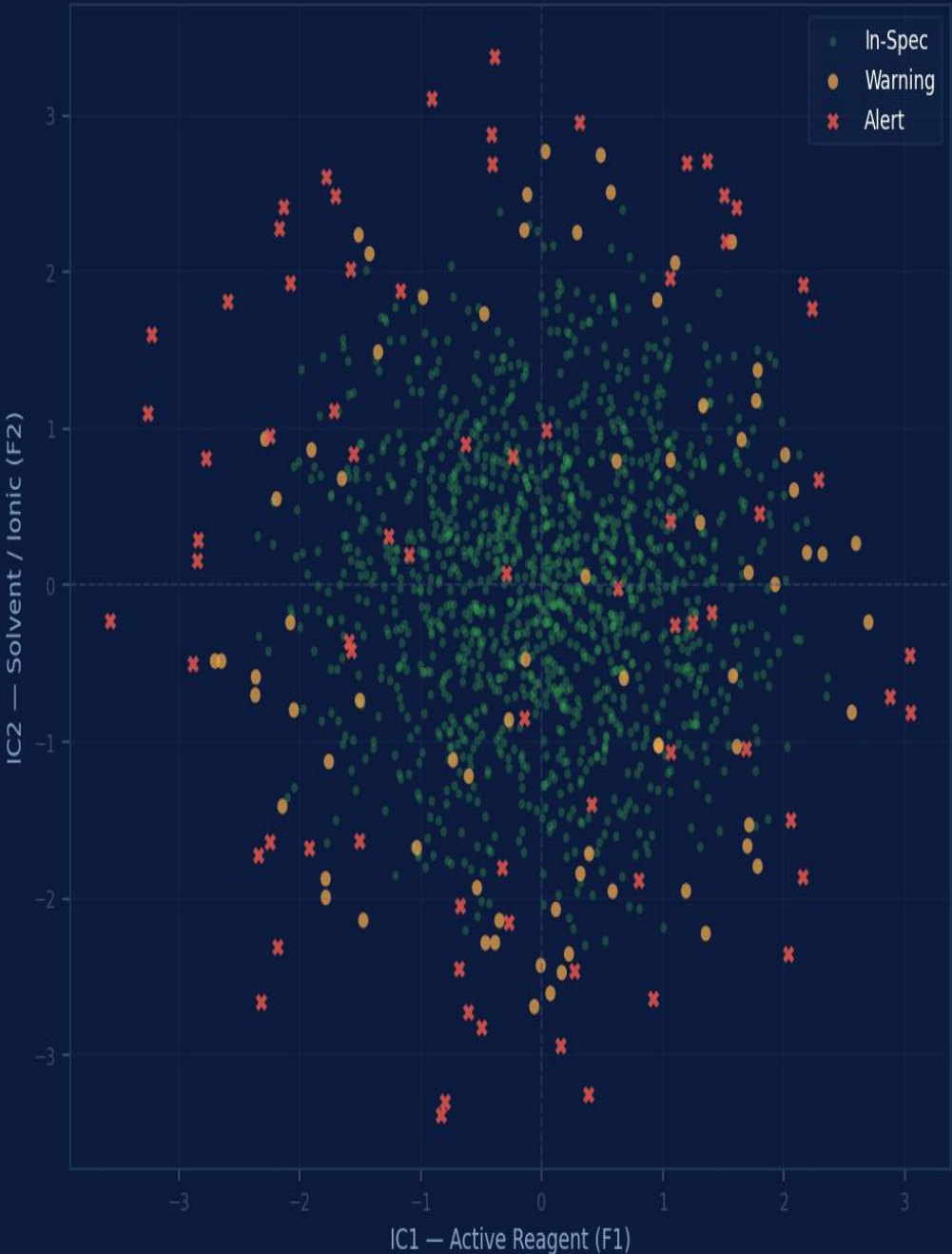
"The batch failed.
But which cause?"

FastICA · k=3

StandardScaler

Mixing Matrix

IC Distance



1,500

Batch Records

3

Hidden Sources (ICs)

90.0%

In-Spec Rate

7 iter

ICA Convergence



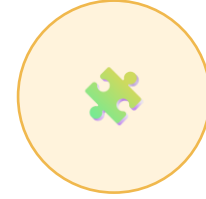
Symptoms, Not Causes

Viscosity out of spec → corrective action: review mixing. Two batches later, the same problem returns from a different direction. Active concentration this time. Then pH. The system was not wrong — it was answering the wrong question.



Eight Sensors, Three Drivers

pH, viscosity, conductivity, density, turbidity, temperature, time, active concentration — eight measurements. Three independent process drivers that none of them observes directly: reagent dosage (F1), solvent quality (F2), agitation kinetics (F3).



The Mixing Problem

When F1 (reagent dosage) drifts, viscosity + density + active concentration all shift together — one event, three non-conformance reports. ICA reverses the mix. It reads the 8 sensors and returns the 3 causes. One event, one corrective action.

ICA does not detect that something is wrong. It tells you what is wrong — and which sensor to move to fix it.

SENSOR BY SENSOR (SPC)

Viscosity: 162 cP → OUT OF SPEC

pH: 7.11 → borderline

Active conc: 103 mg/mL → elevated

→ 3 corrective actions written

→ Root cause: unknown

→ Engineer reviews mixing procedure

→ Mixing was fine — reagent was the issue

→ Same batch failure next month



ICA ROOT CAUSE SPACE

8 sensors → 3 independent components

IC1 = Active Reagent Dosage (F1)

IC2 = Solvent / Ionic Balance (F2)

→ IC3 = Agitation & Kinetics (F3)

→ IC dist = 3.1 — direction: IC1 axis

→ Root cause: reagent dosage event

→ Check lot certificate + dosing pump

→ One action. Correct source. Done.

PAT

In-Line Analyzer

ph_final · conductivity_ms_cm

pH electrode + conductivity cell at batch end — both reflect ionic balance of the solvent system

HPLC

Quality Lab / LIMS

active_conc_mg_ml · density_g_ml · viscosity_cp

Post-batch chromatography for active ingredient · Oscillating U-tube density · Rotational viscometer

DCS

Batch Execution / RTD

reaction_temp_c · reaction_time_min

RTD at reactor jacket logging peak temperature · MES start/stop timestamps for batch duration

PAT

In-Line Nephelometer

turbidity_ntu

Optical turbidity sensor — reflects mixing efficiency and particle size distribution from agitation kinetics

True sources f1, f2, f3 are included for educational validation only — FastICA never sees them during training. In production, only the 8 sensor columns are available.

"PCA finds directions of maximum variance. ICA finds directions of maximum independence. In a real process, what's driving quality is almost never just variance — it's hidden causes."

Independence, Not Variance

PCA groups what moves together (variance). ICA separates what happens for different reasons (independence). Reagent dosage drift has nothing to do with solvent quality variation — they are orthogonal events. ICA's objective matches that reality.

The Mixing Matrix Is the Map

The output is not just a score. It is an 8×3 table that says: when IC1 drifts, expect density (−0.971), active conc (−0.869), and viscosity (−0.866) to co-deviate. That table is the bridge between the model and the corrective action.

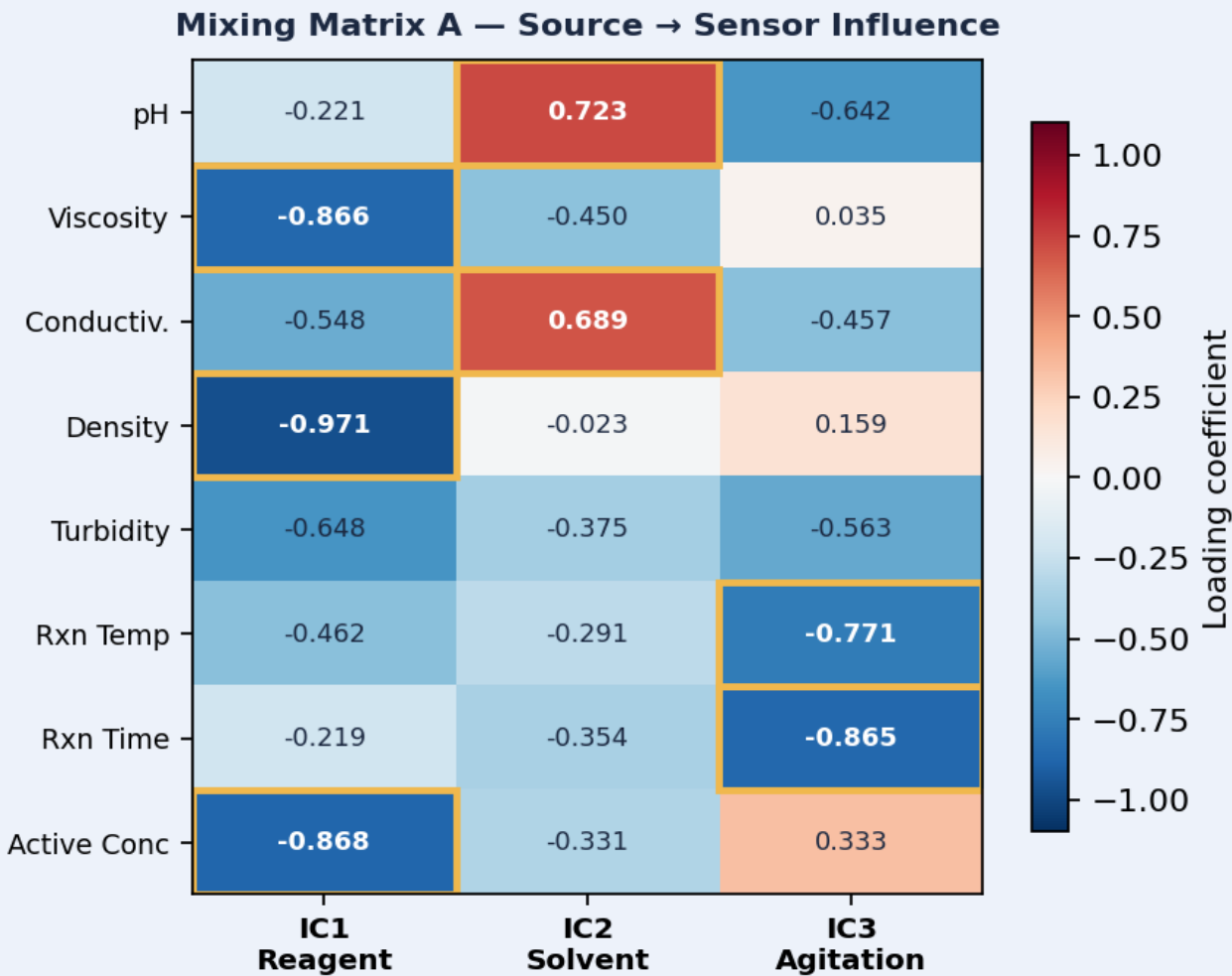
ICA vs PCA: Source Recovery

ICA IC1→F1: $r=0.750$. IC2→F2: $r=0.672$. IC3→F3: $r=0.821$. PCA PC1→F2: $r=0.578$ (partial mix). PCA PC2→F1: $r=0.804$. PCA finds variance directions — they are mixtures of sources. ICA finds sources — one per root cause.

Root Cause, Not Anomaly Flag

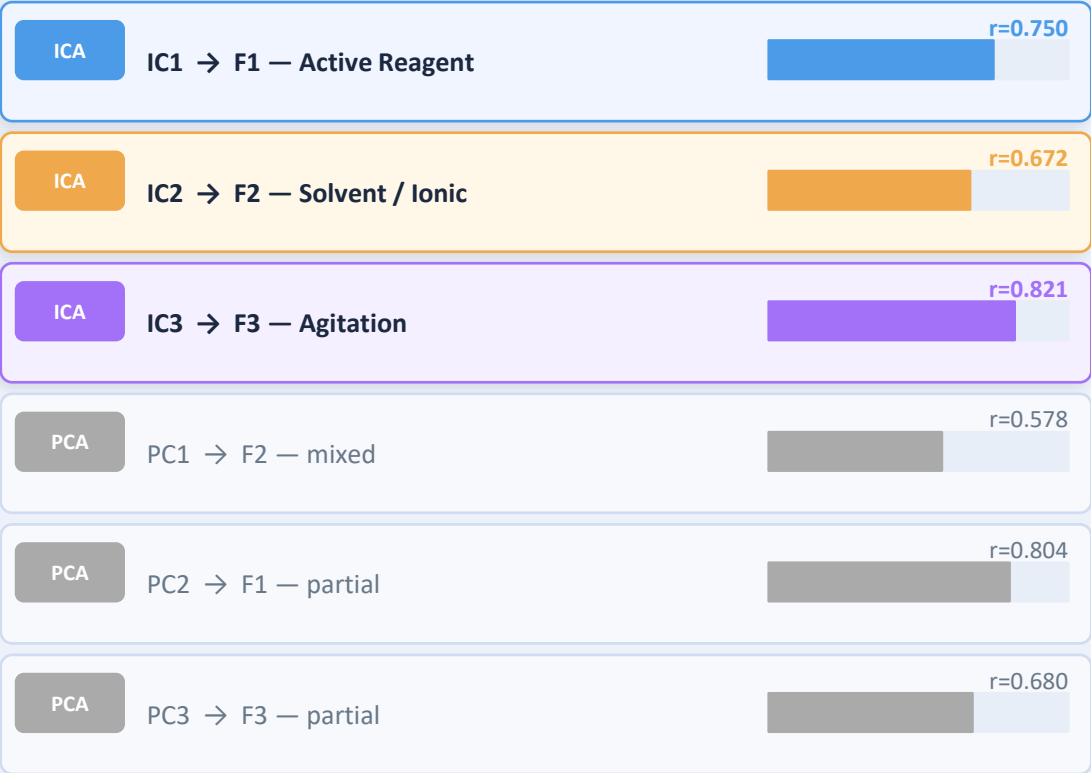
DBSCAN says: this batch doesn't belong. PCA says: IR = 0.68. ICA says: this batch failed along IC3 — check agitator speed and jacket cooling flow. Three tools, three different answers to three different questions.

Mixing Matrix — Source → Sensor Influence

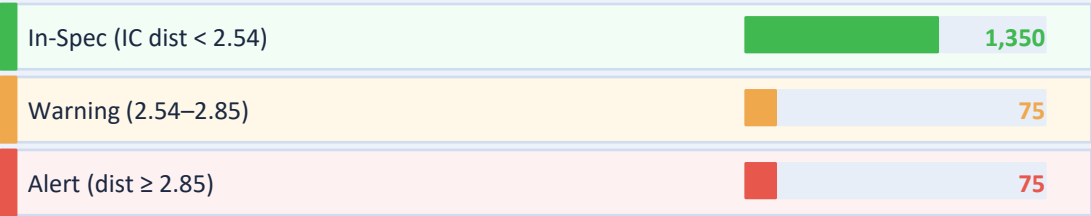


3 PCs = 94.4% variance · FastICA in 7 iterations · IC1→F1: 0.75 · IC2→F2: 0.67 · IC3→F3: 0.82

ICA vs PCA — Source Recovery



Batch Quality Distribution



IC1	Active Reagent Dosage — F1 Dominant sensors: Density (−0.971) · Active Conc (−0.869) · Viscosity (−0.866) Trigger: IC1 distance elevated · IC2 and IC3 within normal range	→	Review batch certificate for reagent purity. Calibrate dosing pump. Verify reagent lot traceability. Check raw material release results before next batch.
IC2	Solvent Quality / Ionic Balance — F2 Dominant sensors: pH (+0.723) · Conductivity (+0.689) Trigger: IC2 distance elevated · pH and conductivity co-deviating in same direction	→	Review water quality — conductivity of purified water feed. Check buffer preparation procedure. Inspect ion exchange resin saturation. Verify solvent lot specification.
IC3	Agitation & Mixing Kinetics — F3 Dominant sensors: Rxn Time (−0.865) · Rxn Temperature (−0.772) · Turbidity (−0.563) Trigger: IC3 distance elevated · Batch runs hotter/faster or cooler/slower than normal	→	Inspect agitator impeller for wear or fouling. Verify agitator speed control loop setpoint. Check cooling jacket flow and temperature control. Review mixing time protocol.
IC1+IC3	Multiple Sources — Reagent + Agitation Event Dominant sensors: All sensors from IC1 and IC3 co-deviating simultaneously Trigger: IC1 and IC3 both elevated · IC dist from combined sources > 3.5	→	Do not write sensor-level corrective actions. This is a process interaction event. Conduct multi-variable root cause analysis. Check if a reagent dosage change altered the optimal mixing setpoint.

A — Reagent Drift (IC1)	
IC1 = 3.10 ▲	
IC2 = 0.33	
IC3 = 0.51	
ph_final	7.02
viscosity_cp	185 cP ▲
conductivity	5.9 mS/cm
density	1.24 g/mL ▲
turbidity	24 NTU
active conc	118 mg/mL ▲

IC dist = 3.1

IC1 Alert — Reagent Event

B — Solvent Event (IC2)	
IC1 = 0.42	
IC2 = 2.87 ▲	
IC3 = 0.58	
ph_final	7.48 ▲
viscosity_cp	151 cP
conductivity	8.2 mS/cm ▲
density	1.20 g/mL
turbidity	23 NTU
active conc	101 mg/mL

IC dist = 2.9

IC2 Alert — Solvent Event

C — Agitation Failure (IC3)	
IC1 = 0.38	
IC2 = 0.61	
IC3 = 3.26 ▲	
ph_final	7.01
viscosity_cp	148 cP
conductivity	5.8 mS/cm
density	1.19 g/mL
turbidity	34 NTU ▲
active conc	98 mg/mL

IC dist = 3.3

IC3 Alert — Kinetics Event

Three batches · same IC distance ≈ 3.0 · same severity · same urgency. Different directions in IC space $\rightarrow$ reagent lot, solvent preparation, agitator failure. Three completely different corrective actions. The distance says how much. The direction says what.

Same IC distance. Three different causes. Three different corrective actions.

Eight sensors were talking.

Three of them were the same conversation.

The mixing matrix told you which three. The IC distance told you which batch.

Every corrective action that cannot name a root cause is just a description of a symptom.

Download. Run it on your own batch quality data.

Change k . Add sensors. Re-label your ICs with your own process knowledge.

Because every corrective action that doesn't name a root cause is just a description.

